

PRACTICAL NO.4

Finding area,perimeter,midpoint of line segment or centroid of 2D objects.point of intersections of two objects and angles between them etc.

Q.1)Area

```
In [1]: from sympy.geometry import *
```

```
In [3]: A,B,C=Point(0,0),Point(4,0),Point(2,3)
t=Triangle(A,B,C)
t.area
```

```
Out[3]: 6
```

```
In [4]: p=Polygon(Point(0,0),Point(4,0),Point(5,3),Point(1,3))
p.area
```

```
Out[4]: 12
```

Q.2)Perimeter

```
In [7]: t.perimeter
```

```
Out[7]: 4 + 2√13
```

```
In [8]: p.perimeter
```

```
Out[8]: 2√10 + 8
```

Q.3)Midpoint

```
In [10]: A,B=Point(1,2),Point(5,6)
seg=Segment(A,B)
seg.midpoint
```

```
Out[10]: Point2D(3, 4)
```

Q.4)Centroid of 2D objects

```
In [11]: t.centroid
```

```
Out[11]: Point2D(2, 1)
```

```
In [12]: p.centroid
```

```
Out[12]: Point2D(5/2, 3/2)
```

Q.5)Intersection of two lines.

```
In [13]: l1=Line(Point(0,0),Point(4,4))
         l2=Line(Point(0,4),Point(4,0))

         l1.intersection(l2)
```

Out[13]: [Point2D(2, 2)]

Q.6)Intersection of Line and Circle

```
In [24]: c=Circle(Point(0,0),5)
         l=Line(Point(-6,0),Point(6,0))

         l.intersection(c)
```

Out[24]: [Point2D(-5, 0), Point2D(5, 0)]

Q.7)Angle between two lines

```
In [25]: l1=Line(Point(0,0),Point(1,0))
         l2=Line(Point(0,0),Point(1,1))

         l1.angle_between(l2)
```

Out[25]: $\frac{\pi}{4}$

Q.8)Distance between objects

```
In [26]: Point(1,2).distance(Point(4,6))
```

Out[26]: 5

Q.9)Distance between Point and Line

```
In [27]: l=Line(Point(0,0),Point(4,0))
         p=Point(2,3)

         l.distance(p)
```

Out[27]: 3

In []: